Econ 695 - Midterm 2 Solutions

November 11, 2025

Date and Time: 8:00-9:15 AM, November 12th, 2025

Instructions: Write legibly and keep your answers **short and clear**. Most questions can be answered in less than half a page. Read the hints carefully — addressing the points in hints can help you get partial credits.

Question 1 Potential Outcomes (35 points; 15 minutes)

Let D_i denote an endogenous choice of unit i that could be either 0 or 1. For units with $D_i = 1$, we observe outcomes $Y_i = Y_i(1)$, and for individuals with $D_i = 0$, we observe $Y_i = Y_i(0)$. Each unit can be characterized by her potential outcomes - $\{Y_i(0), Y_i(1)\}$, and the treatment effect of D_i on outcome Y_i for that unit is defined as $Y_i(1) - Y_i(0)$. In the past few weeks we use the potential outcomes framework in several settings. Let's revisit the identifying assumptions in the models below.

(a) (15 points) Suppose an analyst wants to study the causal impact of smoking on health outcomes. Define $D_i = 1$ if individual i is smoking, and Y_i as her health outcome. Let $\bar{Y}_1$ be the sample mean among smokers, and $\bar{Y}_0$ denote the sample mean among non-smokers. Is $\bar{Y}_1 - \bar{Y}_0$ a consistent estimate for the average treatment effect on the treated $E[Y_i(1) - Y_i(0)|D_i = 1]$?

Hint: $\bar{Y}_0 \xrightarrow{p} E[Y_i|D_i = 0] = E[Y_i(0)|D_i = 0]$.

- Solution:
  - (5 points if answer No) No unless $E[Y_i(0)|D_i = 0] = E[Y_i(0)|D_i = 1]$.
  - (5 points) $\bar{Y}_1 - \bar{Y}_0$ converges in probability to $E[Y_i|D_i = 1] - E[Y_i|D_i = 0]$, and we have

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0] = E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0].$$

- (5 points on selection bias) In contrast, the ATT is defined as $E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 1]$, and

$$ATT = \underbrace{E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{(a)}$$
$$+ \underbrace{E[Y_i(0)|D_i = 0] - E[Y_i(0)|D_i = 1]}_{\text{selection bias}}$$

We have (a) = ATT iff the selection bias is zero, that is $E[Y_i(0)|D_i = 0] = E[Y_i(0)|D_i = 1]$. Without this estimate, $\bar{Y}_1 - \bar{Y}_0 \xrightarrow{p} (a)$ is not a consistent estimate of ATT.

(b) (10 points) The same analyst in (a) realizes the problems you point out in (a), and designs an experiment instead. Some individuals are assigned to a treatment group ($Z_i = 1$) who will attend a consultation with a doctor specialized in lung health, while others are assigned to a control group without any intervention. Under what assumptions is Z_i (treatment group assignment) a valid instrument for D_i (smoking)?

Hint: discuss how Z_i is (un)related to $D_i(0), D_i(1)$ and whether $Y_i = Y_i(D_i, Z_i)$ or $Y_i(D_i)$.

- Solution:
  - (2 points) A valid instrument needs to be relevant for D_i. $cov(D_i, Z_i) \neq 0$. Or in the experiment, there should be some individuals with $D_i(1) < D_i(0)$. If $D_i(1) \equiv D_i(0)$, being assigned to a treatment group does not affect behavior at all.
  - (5 points) The treatment status should be randomly assigned and therefore uncorrelated with unobserved selection into/out of smoking:

    $$Z_i \perp (D_i(0), D_i(1)).$$

    Note: give full 5 points if a student writes $Z_i \perp (D_i(0), D_i(1))$.
  - (3 points) The instrument should not have any direct impact on health. The exclusion restriction can be expressed as $Y_i(d, z) = Y_i(d)$, and in combination with the random assignment assumption, we have:

    $$Z_i \perp (D_i(0), D_i(1), Y_i(0), Y_i(1)).$$

We can also interpret the difference-in-differences model via potential outcomes. Consider the classic Card and Krueger (1994) minimum wage study. Let i denote a restaurant that is either in NJ or in PA (but not both). $t = 1$ denotes the period before NJ raised minimum wage, and $t = 2$ denotes the period after the change. Define $NJ_i = 1$ if restaurant i is in NJ, and $Post_t = 1$ if $t = 2$ (after the minimum wage increase). The treatment variable is defined as $D_{it} = NJ_i \times Post_t$, which equals to 1 if the restaurant i is in NJ, **and** $t = 2$. The outcome of interest is full-time employment Y_{it}, and they estimate the difference-in-differences regression:

$$Y_{it} = \beta_0 + \beta_1 NJ_i + \beta_2 Post_t + \beta_3 D_{it} + \epsilon_{it}.$$

The realized outcomes can be written as $Y_{it} = Y_{it}(D_{it})$.

(c) (10 points) State the parallel trend assumption in words (what it implies for employment changes in NJ vs. PA), and then express it using potential outcomes $Y_{it}(0), Y_{it}(1)$.

- Solution:
  - (5 points for the narrative) The parallel trend assumption is that the change in employment in NJ would have been the same as the change in PA if there had not been a minimum wage increase. (also ok if mention: "employment on parallel trajectories...")
  - (5 points for the potential outcome expression) Using potential outcomes, the parallel trend assumption can be written as:

$$E[Y_{i2}(0) - Y_{i1}(0)|NJ_i = 1] = E[Y_{i2}(0) - Y_{i1}(0)|NJ_i = 0]$$

  or $E[Y_{i2}(0) - Y_{i1}(0)|NJ] = E[Y_{i2}(0) - Y_{i1}(0)|PA]$.

Question 2 Panel Data (35 points; 20 minutes)

Suppose you are given data from surveys at the Twins Festival. There are N families and each family has a pair of identical twins. Let i index a family and $t = 1, 2$ denote the twin #1 and #2. The goal is to use this panel data at (i, t) or family $\times$ twin level to estimate the returns to education. In class we have seen the following regression table. The dependent variable across all columns is the earning of twin t from family i, denoted by Y_{it}. We define the terms:

- S_{it}: own education of twin t in family i.
- $S_{i(-t)}$: sibling education (the other twin's in the same family i). For example, for twin $t = 1$ in family i, $S_{i(-t)} = S_{i2}$ — education of his/her identical twin in the family.
- $\bar{S}_i$: mean education of the twins in family i, $\bar{S}_i = \frac{S_{i1}+S_{i2}}{2}$.
- α_i : fixed effect of family i.

Suppose the causal model is

$$Y_{it} = \beta_0 + \beta_1 S_{it} + \alpha_i + \epsilon_{it} \tag{1}$$

Alternative Estimates of Return to Education Using Twins Sample

	OLS	OLS with Family Dummies	OLS with Mean Educ. Of Pair	OLS with Educ. Of Sibling
Own Education	0.0802	0.0678	0.0678	0.0749
	(0.0071)	(0.0131)	(0.0197)	(0.0106)
Mean Educ. of Pair	--	--	0.0143	--
			(0.0212)	
Sibling Education	--	--	--	0.0072
				(0.0106)
Father's Education	--	--	--	--

where β_1 represents causal effect of education on earnings. The table below shows several estimates of β_1 in sample using different specifications. Note each regression controls for a constant (i.e. β_0), but the estimated intercept is not shown in the table.

(a) (9 points) The first column of the table corresponds to regression: $Y_{it} = \hat{\beta}_0^{OLS} + \hat{\beta}_0^{OLS} S_{it} + \hat{u}_{it}$. Express the sample regressions that correspond to columns 2-4 in the table.
Hint: for the fixed-effect model (2nd column), you could use $\hat{\alpha}_i$ or a set of dummies for families $\{D_{ji} : j = 1...N\}$ with $D_{ji} = 1[j = i]$.

- Solution: 3 points for each regression. (ok if the student forget the hats on coefficients.)

  - Column 2 – fixed effect regression:

$$Y_{it} = \hat{\beta}_0^{FE} + \hat{\beta}_1^{FE} S_{it} + \hat{\alpha}_i + \hat{\epsilon}_{it}$$

$$\text{OR } = \hat{\beta}_0^{FE} + \hat{\beta}_1^{FE} S_{it} + \sum_{j=2}^{N} D_{ji}\hat{\alpha}_j + \hat{\epsilon}_{it}$$

    either specification works.

  - Column 3 – control function:

$$Y_{it} = \hat{\beta}_0^{CF} + \hat{\beta}_1^{CF} S_{it} + \hat{\beta}_2^{CF} \bar{S}_i + \hat{\nu}_{it}$$

  - Column 4 - another control function:

$$Y_{it} = \hat{\delta}_0^{CF} + \hat{\delta}_1^{CF} S_{it} + \hat{\delta}_2^{CF} S_{i(-t)} + \hat{\nu}_{it}$$

(b) (10 points) What assumption do you need to be able to interpret $\hat{\beta}_1^{FE}$ in column 2 (controlling for family dummies) as a **causal** estimate of the return to education? *Hint: think about the error term ϵ_{it} in causal model (1) and how it is (un)related to $(S_{i1}, S_{i2}, \alpha_i)$.*

- Solution: Strict exogeneity. Given causal model (1), the fixed-effect estimator $\hat{\beta}_1^{FE} \xrightarrow{p} \beta_1$ if the errors satisfy:

$$\forall \boldsymbol{s} \in \mathbf{\{1, 2\}} : E[\epsilon_{is}|S_{i1}, S_{i2}, \alpha_i] = 0.$$

  - ok to say the residuals and $(S_{i1}, S_{i2}, \alpha_i)$ are uncorrelated rather than mean independent.
  - take 4 points off if they only state the **contemporary** independence/uncorrelatedness, e.g., $E[\epsilon_{it}|S_{it}] = 0$.

(c) (10 points) In column 4, the estimated coefficient on own education is 0.0749 rather than 0.0678 in the previous two columns. Are we getting a different estimate for the return to education? Explain why or why not. Hint: compare the regressions you write down in (a) for column 3 and 4.

- Solution: No. (3 points for the correct answer)
  - (3 points for pointing out...) We get the same estimate by taking the difference: $\hat{\delta}_1^{CF} - \hat{\delta}_2^{CF} = 0.0678$ too.
  - (4 points for explanation) The regression for column 4 is equivalent to the regression for column 3. To see this, note $\bar{S}_i = \frac{S_{i1}+S_{i2}}{2}$. Rearranging the terms we have:

$$\begin{aligned} Y_{it} &= \hat{\beta}_0^{CF} + \hat{\beta}_1^{CF} S_{it} + \hat{\beta}_2^{CF} \frac{S_{i1} + S_{i2}}{2} + \hat{\nu}_{it} \\ &= \hat{\beta}_0^{CF} + (\hat{\beta}_1^{CF} + \frac{\hat{\beta}_2^{CF}}{2}) S_{it} + \frac{\hat{\beta}_2^{CF}}{2} S_{i2} + \hat{\nu}_{it} \end{aligned}$$

    Given that $\hat{\nu}_{it}$ is orthogonal to the regressors, we will get $\hat{\delta}_1^{CF} = \hat{\beta}_1^{CF} + \frac{\hat{\beta}_2^{CF}}{2}$ and $\hat{\delta}_2^{CF} = \frac{\hat{\beta}_2^{CF}}{2}$, so $\hat{\delta}_1^{CF} - \hat{\delta}_2^{CF} = \hat{\beta}_1^{CF}$.

(d) (6 points) If I take the difference in earnings between twins in each family, and regress it on the difference in education,

$$Y_{i2} - Y_{i1} = \beta_1^{FD} \times (S_{i2} - S_{i1}) + \epsilon_i$$

What would be the estimate $\hat{\beta}_1^{FD}$ in this first-difference regression? Provide a brief explanation.

- Solution:

- (3 points for the correct answer) $\hat{\beta}_1^{FD} = 0.0678$.
- (3 points) Since $t = 1, 2$ in this panel data, first-difference estimators are the same as within or fixed effect estimators.
  By Frisch-Waugh on the fixed-effect model, one can obtain $\hat{\beta}_1^{FE}$by regressing earning on own education residualized over family dummies:

$$\hat{\beta}_1^{FE} = \frac{\sum_{it} Y_{it}(S_{it} - \bar{S}_i)}{\sum_{it}(S_{it} - \bar{S}_i)^2} = \frac{\sum_{it}(Y_{it} - \bar{Y}_i)(S_{it} - \bar{S}_i)}{\sum_{it}(S_{it} - \bar{S}_i)^2}$$

  That is, this is equivalent to running a regression of $Y_{it} - \bar{Y}_i$ on $S_{it} - \bar{S}_i$.
  Note $Y_{i2} - \bar{Y}_i = \frac{Y_{i2} - Y_{i1}}{2}$ and $S_{i2} - \bar{S}_i = \frac{S_{i2} - S_{i1}}{2}$. The model above is equivalent to the first-difference model (multiply both sides by 2). Therefore, under T=2, we have $\hat{\beta}_1^{FE} = \hat{\beta}_1^{FD}$.

Question 3 Local Average Treatment Effects (30 points; 30 minutes)

Consider a randomized experiment to encourage children to be vaccinated. The vaccination requires 2 shots, administered in two consecutive weeks. Define $Z_i = 1$ if a kid i is randomly assigned to the treatment group with encouragement, and $Z_i = 0$ if assigned to the control group without encouragement.

Suppose that we know that the population consists of 4 *types*:
- NT = never takers: these children never take either shot, regardless of whether they are in the control group ($Z_i = 0$) or treatment group ($Z_i = 1$);
- AT = always takers: these children always take both shots, regardless of whether they are in the control group or treatment group;
- $CP1$ = "partial compliers": if assigned to treatment these children take the first shot but not the second; if assigned to the control group they take no shots;
- $CP2$ = "full compliers": if assigned to the treatment group these children take both shots ; if assigned to the control group they take no shots.

Suppose the probabilites of these groups in the experimental population are $P(NT)$, $P(AT)$, $P(CP1)$, $P(CP2)$, and that there are *NO DEFIER*S.

Define 3 potential outcomes for each child i:
$Y_i(0)$: indicator for whether child i gets the disease if she has NO SHOTS
$Y_i(1)$: indicator for whether child i gets the disease if she has 1 SHOT
$Y_i(2)$: indicator for whether child i gets the disease if she has BOTH SHOTS

Define two dummy variables:

$$\begin{aligned} D_{1i} &= 1 \text{ if i gets 1st shot ONLY} \\ D_{2i} &= 1 \text{ if i gets BOTH shots} \end{aligned}$$

After random assignment you observe y_i which is a dummy variable indicating that i got the disease (or not), and the number of shots received by each child, for all children in both the treatment group and the control group.

(a) (6 points) Express the four types by $(D_{1i}(0), D_{1i}(1), D_{2i}(0), D_{2i}(1))$, in which $D_{1i}(0) = 1$ if i gets the first shot only when she is assigned to the control group $Z_i = 0$, and $D_{1i}(1) = 1$ if she gets the first shot only when she is assigned to the treatment group. The partial compliers (CP1) are characterized by: $D_{1i}(0) = D_{2i}(0) = 0$ if in the control group, and $D_{1i}(1) = 1$, $D_{2i}(1) = 0$ if in the treatment group. Complete the table below (filling in 0 or 1 in empty cells):

Solution: 0.5 point for each correct entry in blue

	If $Z_i = 0$		If $Z_i = 1$	
	$D_{1i}(0)$	$D_{2i}(0)$	$D_{1i}(1)$	$D_{2i}(1)$
NT	0	0	0	0
AT	0	1	0	1
$CP1$	0	0	1	0
$CP2$	0	0	0	1

(b) (4 points) Show that $E[D_{2i}|Z_i = 1] = P(AT) + P(CP2)$. *HINT: by the law of iterated expectations:*

$$\begin{aligned} E[D_{2i}|Z_i = 1] &= E[D_{2i}|NT,\, Z_i = 1] \times P(NT \,|\, Z_i = 1) \\ &\quad + E[D_{2i}|AT,\, Z_i = 1] \times P(AT \,|\, Z_i = 1) \\ &\quad + E[D_{2i}|CP1,\, Z_i = 1] \times P(CP1 \,|\, Z_i = 1) \\ &\quad + E[D_{2i}|CP2,\, Z_i = 1] \times P(CP2 \,|\, Z_i = 1) \end{aligned}$$

Simply the expression above using random assignment assumption. Which types have $D_{2i} = 1$ when assigned to the treatment group (according to your table in (a))?

- Solution:
  - Following the hint, by L.I.E., $E[D_{2i}|Z_i = 1] = E_g[E[D_{2i}|Z_i = 1, g]] = E_g[E[D_{2i}(1)|Z_i = 1, g]]$
  - under random assignment, $Pr(g|Z_i = 1) = Pr(g)$ for g = AT/NT/CP1/CP2. And $Z_i \perp (D_{2i}(0), D_{2i}(1))$. So

$$E[D_{2i}|Z_i = 1] = \sum_g Pr(g) \times E[D_{2i}(1)|g]$$

- Based on the potential treatments in (a), $D_{2i}(1) = 1$ only for AT and CP2. For other types, $D_{2i}(1) = 0$. Therefore, $E[D_{2i}|Z_i = 1] = Pr(AT) + Pr(CP2)$.

(c) (5 points) Show that $E[Y_i|AT, Z_i = 0] = E[Y_i(2)|AT]$.

Hint: think about the behavior of always takers if assigned to the control group. What is the observed outcome for these children?

$$Y_i = (1 - D_{1i} - D_{2i}) \times Y_i(0) + D_{1i} \times Y_i(1) + D_{2i} \times Y_i(2) \quad (*)$$

- Solution:
  - (2 points) Conditional on $Z_i = 0$, always takers have $D_{2i} = D_{2i}(1) = 1$ based on our table in (a).
  - (1 point) $E[Y_i|AT, Z_i = 0] = E[Y_i(2)|AT, Z_i = 0]$ given that $D_{2i} = 1$.
  - (2 points) By random assignment, we can then drop the condition $Z_i = 0$:

$$E[Y_i|AT, Z_i = 0] = E[Y_i(2)|AT].$$

(d) (10 points) Develop expressions of $E[Y_i|Z_i = 0]$ and $E[Y_i|Z_i = 1]$ in terms of the fractions of the 4 types $\{Pr(AT), Pr(NT), Pr(CP1), Pr(CP2)\}$, and expressions of the form: $E[Y_i(0)|g]$, where type $g \in \{AT, NT, CP1, CP2\}$. *Hint: use the law of iterated expectations and the random assignment assumption. Also use the hint (*) for (c).*

- Solution:
  - (2 points) By L.I.E., $E[Y_i|Z_i = 0] = E_g[E[Y_i|Z_i = 0, g]] = \sum_{g \in \{AT,NT,CP1,CP2\}} Pr(g|Z_i = 0) \times E[Y_i|Z_i = 0, g]$.
  - (3 points) By random assignment, $Pr(g|Z_i = 0) = Pr(g)$ and

$$\begin{aligned} E[Y_i|Z_i = 0, AT] &= E[Y_i(2)|Z_i = 0, AT] = E[Y_i(2)|AT] \\ E[Y_i|Z_i = 0, NT] &= E[Y_i(0)|NT] \\ E[Y_i|Z_i = 0, CP1] &= E[Y_i(0)|CP1] \\ E[Y_i|Z_i = 0, CP2] &= E[Y_i(0)|CP2] \end{aligned}$$

    Therefore $E[Y_i|Z_i = 0] = Pr(AT)E[Y_i(2)|AT] + Pr(NT)E[Y_i(0)|NT] + Pr(CP1)E[Y_i(0)|CP1] + Pr(CP2)E[Y_i(0)|CP2]$.
  - (5 points) Similarly, conditional on $Z_i = 1$,

$$\begin{aligned} E[Y_i|Z_i = 1, AT] &= E[Y_i(2)|AT] \\ E[Y_i|Z_i = 1, NT] &= E[Y_i(0)|NT] \\ E[Y_i|Z_i = 1, CP1] &= E[Y_i(1)|CP1] \\ E[Y_i|Z_i = 1, CP2] &= E[Y_i(2)|CP2] \end{aligned}$$

    By LIE and random assignment above, we have $E[Y_i|Z_i = 1] = Pr(AT)E[Y_i(2)|AT] + Pr(NT)E[Y_i(0)|NT] + Pr(CP1)E[Y_i(1)|CP1] + Pr(CP2)E[Y_i(2)|CP2]$.

(e) (5 points) Show that the difference in mean outcomes of the treatment and control groups, $E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]$ can be expressed by the fractions of compliers—$Pr(CP1), Pr(CP2)$ and treatment effects for compliers—$E[Y_i(1)-Y_i(0)|CP1]$, and $E[Y_i(2)-Y_i(0)|CP2]$.

- Solution: based on the results in (d),

$$\begin{aligned} E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] =& Pr(CP1)\left(E[Y_i(1)|CP1] - E[Y_i(0)|CP1]\right) \\ &+ Pr(CP2)\left(E[Y_i(2)|CP2] - E[Y_i(0)|CP2]\right) \end{aligned}$$

  that is, we express the difference as a weighted average of the treatment effect for partial compliers CP1, and the treatment effect for complete compliers CP2.